

ECON 351 · Focused Practice Set

Set 7 of 10

Chapter 9

Insurance Problems

5 questions in this practice set

Questions: Q13, Q14, Q15, Q16, Q17

Screenshots from original HW, sample questions, and mock midterm.

Question 1 of 5 (Practice Q13)

Insurance #1 · HW2 Ex 3

Exercise 3 - Insurance

Tom has the following utility from income:

$$u(I) = \sqrt{x}$$

He bought a house in Brentwood for \$846,400. With probability 25%, high winds will cause \$174,000 in damages to the house. In this case, the house value becomes \$672,400. AAA offers to fully insure the house for a price P .

- a) What is the maximum insurance premium P that Tom is willing to pay?
- b) How much is the expected loss? (also known as the fair insurance premium)
- c) How much is the risk premium? (difference between the max.premium the consumer is willing to pay and the expected loss)

Question 2 of 5 (Practice Q14)

Insurance #2 · HW2 Ex 3b

Exercise 3 - Insurance

Tom has the following utility from income:

$$u(I) = \sqrt{x}$$

He bought a house in Brentwood for \$846,400. With probability 25%, high winds will cause \$174,000 in damages to the house. In this case, the house value becomes \$672,400. AAA offers to fully insure the house for a price P .

- a) What is the maximum insurance premium P that Tom is willing to pay?
- b) How much is the expected loss? (also known as the fair insurance premium)
- c) How much is the risk premium? (difference between the max.premium the consumer is willing to pay and the expected loss)

Question 3 of 5 (Practice Q15)

Insurance #3 · Sample Q18

Question 16:

Using the information from the previous question, how much is the fair insurance premium?

- (A) 174,000
- (B) 72,000
- (C) 130,500
- (D) 43,500
- (E) 168,100

Question 17:

Using the information from Question 15, find Tom's associated risk premium:

- (A) 1,200
- (B) 2,100
- (C) 1,550
- (D) 2,000
- (E) 1,875

Question 18:

Sherman has the following utility from wealth:

$$u(x) = \sqrt{x}.$$

Wealth is the sum of cash and the value of his car. He currently has a Ferrari worth \$230,900 and \$50,000 in cash (risk free). With probability 25%, a paparazzi will cause \$158,400 in damages to the car (in this case, the car value becomes \$72,500). With probability 75% nothing happens, and the car keeps its original value. AAA offers to fully insure the car for a price P . What is the maximum insurance premium P that Sherman is willing to pay? What is the associated risk premium?

Question 19:

Joe's utility from income is given by

$$u(I) = 8\sqrt{I}.$$

He has a house in LA County. With probability $1/4$, the rain will cause a mudslide and damage his house, decreasing its value to \$250,000. With probability $3/4$, no mudslide occurs and the house retains its original \$1,000,000 value. AAA offers to fully insure the house for a price P . What is the maximum insurance premium P that Joe is willing to pay? What is the associated risk premium?

Question 20:

Marcelo has the following utility from income: $u(I) = \sqrt{I}$. He has a \$10,000 car. With probability 25% he will crash his car, and will have to pay \$6,348 to the repair shop to fix it. With probability 75% no crash occurs, and he does not have to repair the car. Progressive Insurance offers to partially insure the car for a price $P=124$. That is, Marcelo pays insurance premium $P=124$, and in case of a crash, Progressive will only pay for half of the repair cost (in case of a crash, Progressive will only pay \$3,174 to Marcelo). If Marcelo buys this partial insurance plan, his expected utility will be:

- (A) 94
- (B) 98
- (C) 96
- (D) 95
- (E) 97

Question 4 of 5 (Practice Q16)

Insurance #4 · Mock Q27

Question 27:

[Show your work] A consumer has \$1,000 in cash. With probability 90% nothing happens (he does not get sick) and he is able to keep all his money. With probability 10% he gets sick and must go to the doctor. The cost of going to the doctor is \$500.

An insurance company offers the following health insurance plan. The consumer must pay upfront a premium $p = \$100$. If the consumer gets sick, the insurance company pays the doctor, but the consumer must pay a \$275 copay.

The consumer's utility from income is given by $u(I) = \sqrt{I}$. Compute the expected utility of the consumer if he purchases this insurance.

- (A) 29.5
- (B) 30.7
- (C) 28.6
- (D) 30
- (E) 29

Question 5 of 5 (Practice Q17)

Insurance #5 · Mock Q25-26

Question 25:

[Show your work] A consumer has the following utility from income: $u(I) = \sqrt{I}$. He bought a car for \$10,000. With probability 75%, a crash will cause \$3,600 in damages to the car (in this case, the car's value becomes \$6,400). AAA offers to fully insure the car for a price P . What is the maximum insurance premium P that the consumer is willing to pay?

- (A) 2,725
- (B) 2,850
- (C) 2,750
- (D) 2,825
- (E) 2,775

Question 26:

[Show your work] Using the information from the previous question, compute the associated risk premium.

- (A) 75
- (B) 125
- (C) 50
- (D) 150
- (E) 25